Experiment Report: `pushup_k3_from_spec27_30k`

Push-Up to k=3 — Three-Level CMS with Proxy Tape at L1/L2

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Abstract

This report documents the `pushup_k3_from_spec27_30k` experiment: a push-up from a k=2 checkpoint (spec27 at step 30K) to k=3, adding a new Level 2 (chunk_size=64) with proxy tape strategy. The run is configured for 100K steps but stopped at step 95,640 (96% complete).

This is the first experiment with **two proxy levels** (L1 and L2), testing the Spec 27 strategy at the next push-up stage. Key findings:

- **Loss**: Starting at 3.84 (post-push-up spike), descends to smoothed 2.53 at 50K, reaching 2.74 at final measurement. Improves substantially over the k=2 source checkpoint.
- **Three-tier gate hierarchy**: L0 theta≈1.0 (active learner), L1 theta bimodal (Blocks 1/2/3 active, Block 0 suppressed), L2 theta <0.05 everywhere (passive retainer).
- **No L1 eta collapse**: Unlike the 130K k=2 run, L1 eta remains >0.93 in all blocks at 95K — the presence of L2 below may prevent L1 suppression.
- **Block 2 anomaly persists**: Block 2's L1 theta reaches 0.81 and its L2 theta is the highest of any block at L2 (0.04–0.05) — consistently the most active proxy learner.
- **Universal alpha floor-pinning**: All three levels hit the 0.80 floor in most blocks by step 50K, except L2 in Blocks 2/3 (0.94–0.96).
- **Throughput**: 1,148 tok/s mean — 3% slower than k=2 (1,186 tok/s), consistent with the additional level.

Contents

1 Experiment Configuration **2**
1.1 Push-Up Mechanics . . . 2

2 Loss Trajectory **3**
2.1 Comparison with k=2 Runs . . . 4

3 Gradient Analysis **4**
3.1 Gradient Spikes . . . 5
3.2 Depth Specialization . . . 6

4 Gate Diagnostics: Three-Level Hierarchy **6**
4.1 Theta: The Learning Rate Hierarchy . . . 6
4.2 Alpha: Universal Floor-Pinning . . . 8
4.3 Eta: No Collapse at k=3 . . . 9
4.4 DGD Delta Norms: L0 Only . . . 10
4.5 Block 2: The Persistent Anomaly . . . 10

5 Output Gradient Norms **11**

6 Throughput **11**

7 Conclusions **12**
7.1 Push-Up to k=3 is Viable 12
7.2 Three-Tier Hierarchy: Exact > Inherited Proxy > Fresh Proxy 12
7.3 No L1 Eta Collapse: The Pressure Relief Hypothesis 12
7.4 Alpha Floor is Load-Bearing at k=3 12
7.5 Next Steps 12

1 Experiment Configuration

Table 1: Configuration. Push-up from k=2 spec27 checkpoint to k=3.

Parameter	**Value**
Architecture	
d_model	512
num_heads	8
n_blocks	4
k / chunk_sizes	3 / [1, 8, 64]
composition	MAG
memory_rule	Titans LMM
Tape Strategy (Spec 27)	
tape_strategies	[exact, proxy, proxy]
tape_multiplier	1
tape_device	GPU
Push-Up	
source checkpoint	`spec27_d512_4b_k2/checkpoints/model.safetensors`
extend_k	3 (from 2)
push_up	true
projection_kind	static
momentum_kind	none
Training	
configured steps	100,000
actual steps	95,640 (96%)
lr / warmup	3e-4 / 500
alpha_floor	[0.8, 0.8, 0.8]
m_norm_max	[100.0, 100.0, 100.0]
seed	42
data	Dolmino-100B

1.1 Push-Up Mechanics

The push-up from k=2 to k=3 works as follows:

- L0 and L1 weights are loaded from the k=2 checkpoint (spec27 at step 30K, loss 2.72).
- L2 is initialized fresh with random weights, chunk_size=64 (fires every 64th token).
- The LR schedule resets to a fresh 100K-step cosine, so L2 trains from scratch alongside the pre-trained L0/L1.
- The loss spikes from 2.72 (checkpoint) to ∼3.84 at step 0 as the randomly initialized L2 disrupts the established L0/L1 dynamics.

The source checkpoint is from the *first phase* (30K) of the spec27_d512_4b_k2 run — the same checkpoint that was extended to 130K in the previous report. This push-up branch tests whether adding a third level is more productive than continuing at k=2.

2 Loss Trajectory

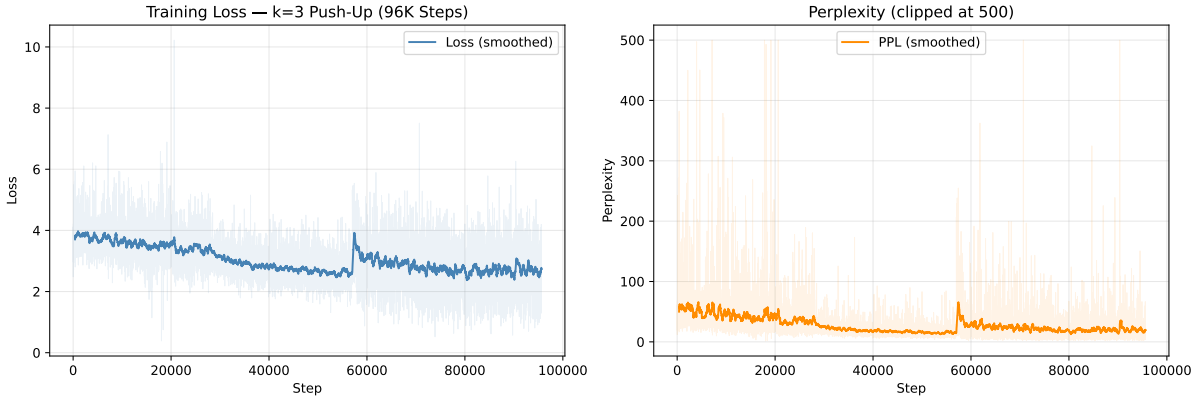

Figure 1: Training loss and perplexity over 96K steps. The initial spike (3.84) from the push-up recovers within ∼5K steps. Loss reaches a minimum around step 50K (smoothed 2.53) before oscillating in the 2.5–2.8 range.

Table 2: Loss at key milestones (last-50 smoothed averages).

Step	Smoothed Loss	Notes
0	3.84	Post-push-up spike
1,000	3.83	Still recovering
5,000	3.64	Recovery underway
10,000	3.75	Temporary plateau
20,000	3.65	Gradual descent
30,000	3.03	Major improvement
50,000	2.53	Best smoothed loss
75,000	2.70	Oscillation
95,640	2.74	Run stopped (final)

The loss trajectory shows a non-monotonic pattern: after the rapid descent to 2.53 at step 50K, the smoothed loss oscillates between 2.5 and 2.8 rather than continuing to improve. The raw final-step loss is 2.29, but the smoothed average is 2.74. This oscillation differs from the k=2 130K run, which showed steady descent to 2.60.

2.1 Comparison with k=2 Runs

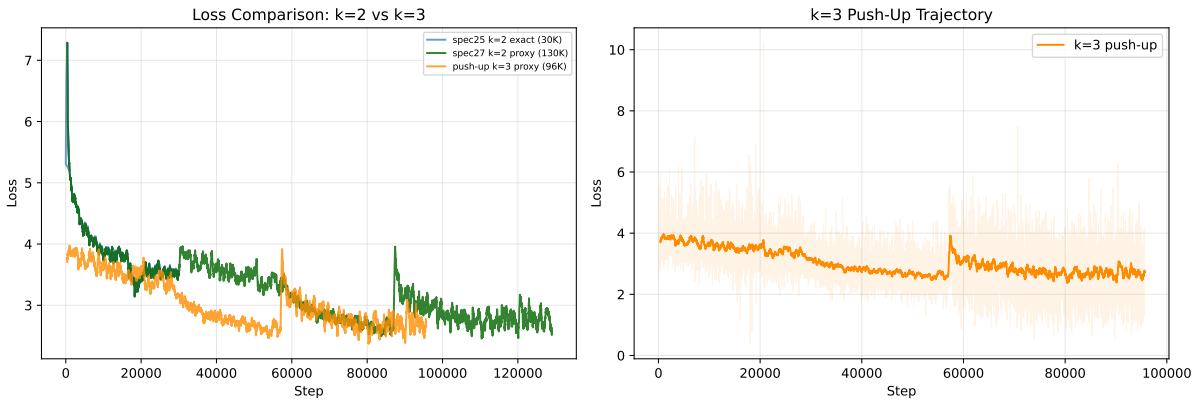

Figure 2: Left: Three-way comparison. The k=3 push-up (orange) starts higher due to the push-up spike but reaches comparable loss to the k=2 130K run by step ∼40K. Right: k=3 trajectory alone.

Table 3: k=3 vs k=2 comparison at matched step counts.

Metric	**k=2 spec25 (30K)**	**k=2 spec27 (130K)**	**k=3 push-up (96K)**
Best smoothed loss	3.53	2.60	2.53
Final smoothed loss	3.53	2.60	2.74
Throughput (tok/s)	1,207	1,186	1,148
Grad spikes (>10)	8	18	9

The k=3 push-up achieves a better *minimum* smoothed loss (2.53) than the k=2 130K run (2.60), but its *final* smoothed loss (2.74) is worse due to late-training oscillation. The k=3 model may benefit from a longer cosine schedule or learning rate adjustment in the final phase.

3 Gradient Analysis

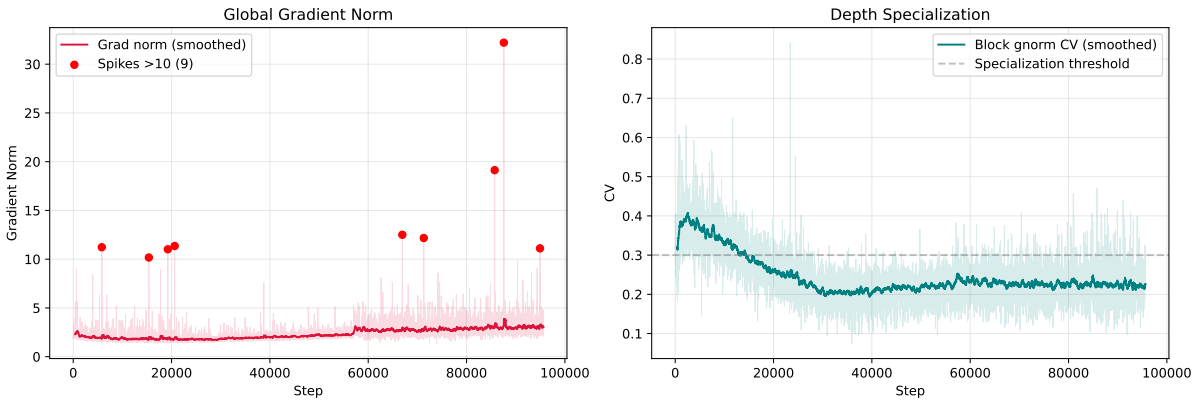

Figure 3: Global gradient norm and block CV. Gnorm settles to ∼2.0–2.5. Block CV drops from 0.38 to 0.20, crossing the specialization threshold around step 15K.

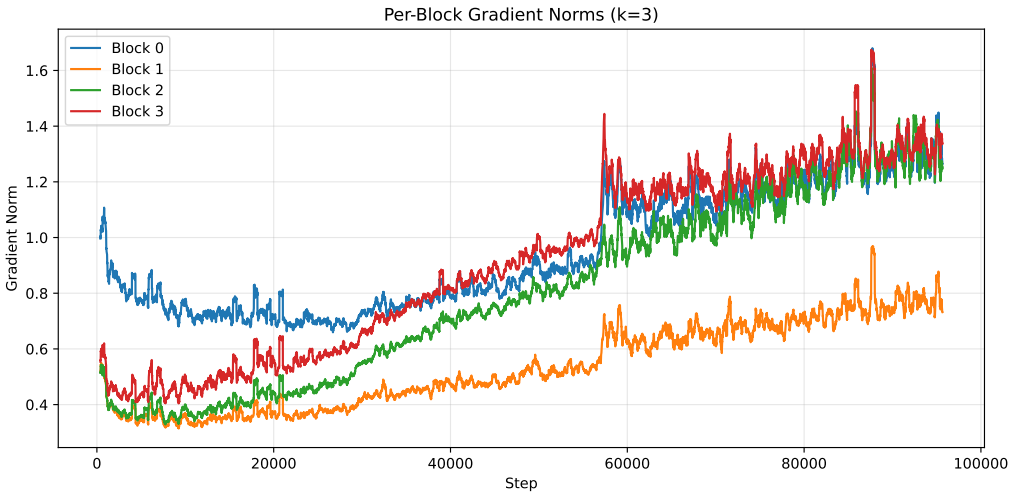

Figure 4: Per-block gradient norms. Block 0 dominates, consistent with all k=2 runs.

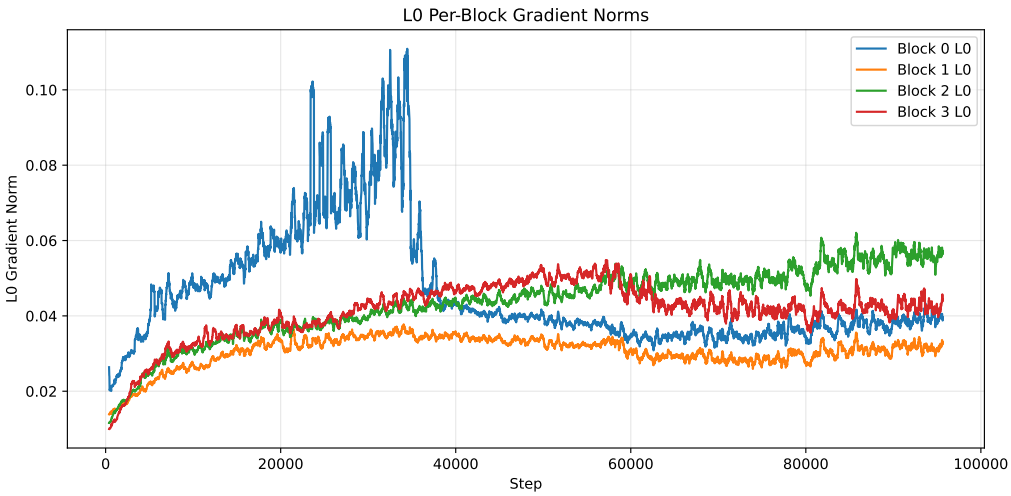

Figure 5: L0 per-block gradient norms.

3.1 Gradient Spikes

Table 4: All 9 gradient spikes (>10).

Step	Grad Norm	Notes
5,816	11.23	Early training
15,384	10.17	Mid training
19,240	11.02	Mid training
20,640	11.35	Mid training
66,936	12.50	Late training
71,296	12.17	Late training
85,720	19.13	Late training
87,584	**32.21**	**Maximum**
94,944	11.11	Near end

With 9 spikes over 96K steps, k=3 has fewer spikes than the k=2 130K run (18 over 130K) on a per-step basis. The largest spike (32.2 at step 87,584) is comparable to the k=2 130K maximum (36.6 at step 117,584). All spikes are contained by gradient clipping.

3.2 Depth Specialization

Table 5: Block CV at milestones.

Step	Block CV
1,000	0.376
10,000	0.338
30,000	0.201
50,000	0.223
75,000	0.227

Block CV starts lower than the k=2 130K run (0.38 vs 0.55) because the k=3 model inherits the pre-homogenized block structure from the k=2 checkpoint (which had already trained for 30K steps with declining CV). The CV stabilizes around 0.20–0.23 from step 30K onward — slightly higher than the k=2 130K final value (0.18), suggesting k=3 provides marginally more reason for block differentiation.

4 Gate Diagnostics: Three-Level Hierarchy

The central finding of this experiment is the emergence of a **three-tier gate hierarchy** under mixed exact/proxy training. Each level occupies a distinct role.

4.1 Theta: The Learning Rate Hierarchy

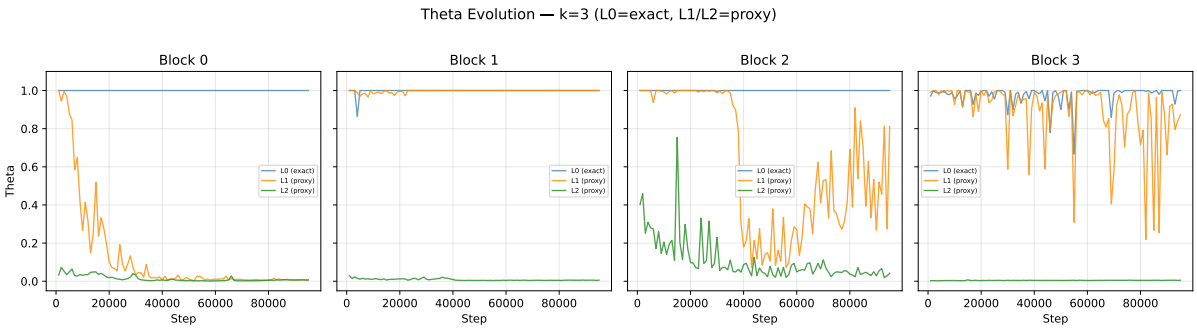

Figure 6: Theta evolution for all three levels across 4 blocks. L0 (blue) saturates to 1.0. L1 (orange) is bimodal: high in Blocks 1/2/3, low in Block 0. L2 (green) stays near zero everywhere.

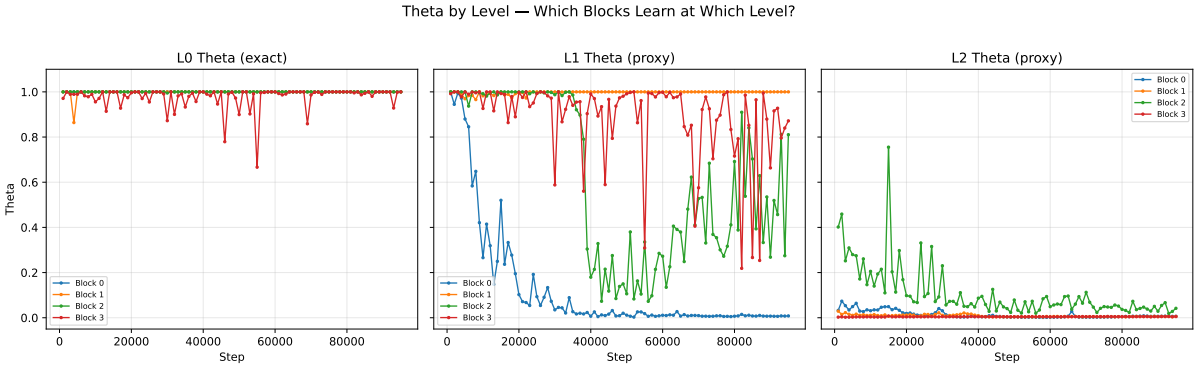

Figure 7: Theta organized by level. Left: L0 theta — all blocks near 1.0. Center: L1 theta — Block 2 reaches 0.81, Block 1 and Block 3 oscillate around 0.5–1.0, Block 0 stays below 0.04. Right: L2 theta — all blocks below 0.05.

Table 6: Theta at step 95K by block and level. The three-tier hierarchy is clear.

Block	**L0 (exact)**	**L1 (proxy)**	**L2 (proxy)**
Block 0	1.000	0.008	0.006
Block 1	1.000	1.000	0.006
Block 2	1.000	0.810	0.042
Block 3	1.000	0.872	0.005

Interpretation: The three-tier pattern reflects gradient fidelity:

- **L0 (exact, $\theta \approx 1.0$)**: Full M trajectory gradients $\rightarrow$ maximum inner-loop LR. Active learner in all blocks.
- **L1 (proxy, bimodal)**: Inherited from k=2 with pre-trained weights. Blocks 1/2/3 maintain high theta from the k=2 phase, suggesting the pre-trained L1 weights provide a strong enough signal even under proxy. Block 0's L1 was already being suppressed in the k=2 run (theta 0.02 at 30K) and remains suppressed.
- **L2 (proxy, $\theta < 0.05$)**: Initialized fresh with no pre-training. Under proxy gradients from the start, L2 never develops a meaningful learning rate. It is a passive retainer, not an active learner.

The key insight: **L1's high theta in Blocks 1/2/3 is inherited, not learned under proxy**. It was trained to theta≈1.0 during the k=2 exact-gradient phase and maintained that state. L2, which never had an exact-gradient phase, cannot bootstrap to high theta under proxy alone. This suggests that the push-up pipeline's value comes from pre-training levels under exact gradients before transitioning to proxy.

4.2 Alpha: Universal Floor-Pinning

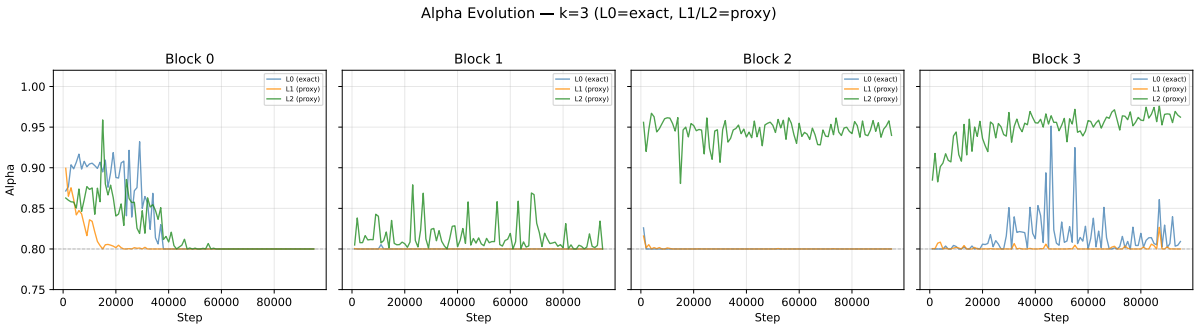

Figure 8: Alpha evolution. All three levels trend toward the 0.80 floor. L2 in Blocks 2/3 stays elevated (0.94–0.96) — these blocks retain more L2 memory.

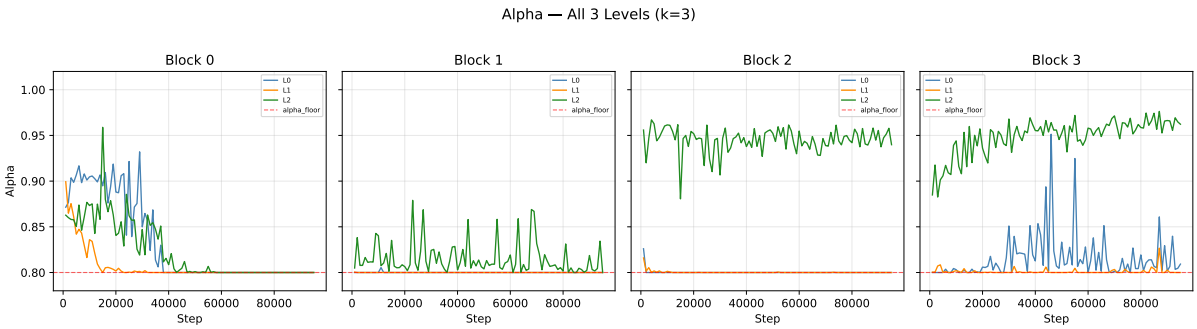

Figure 9: Alpha floor-pinning detail. Most block/level combinations hit the floor by step 50K.

Table 7: Alpha at step 95K. Floor=0.80.

Block	**L0**	**L1**	**L2**
Block 0	0.800	0.800	0.800
Block 1	0.800	0.800	0.800
Block 2	0.800	0.800	0.940
Block 3	0.809	0.800	0.962

The alpha floor (0.80) is saturated across nearly all block/level combinations. This is more aggressive floor-pinning than the k=2 runs, where some L1 alpha values stayed above 0.85. At k=3, the model pushes all levels to maximum forgetting speed allowed by the constraint. L2 in Blocks 2/3 is the exception, retaining alpha≈0.95 — these are the blocks where L2 theta is highest (Block 2: 0.04), suggesting that blocks which learn more at L2 also retain more.

4.3 Eta: No Collapse at k=3

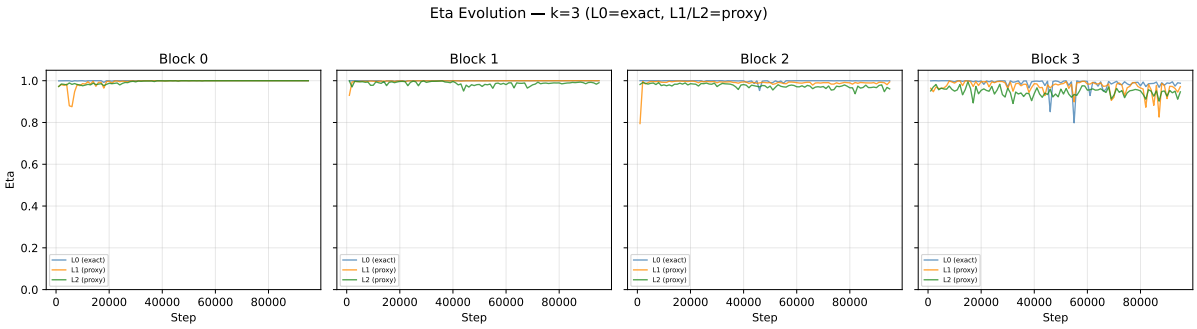

Figure 10: Eta evolution. All three levels maintain high eta (>0.93) across all blocks. No L1 eta collapse is observed, unlike the k=2 130K run.

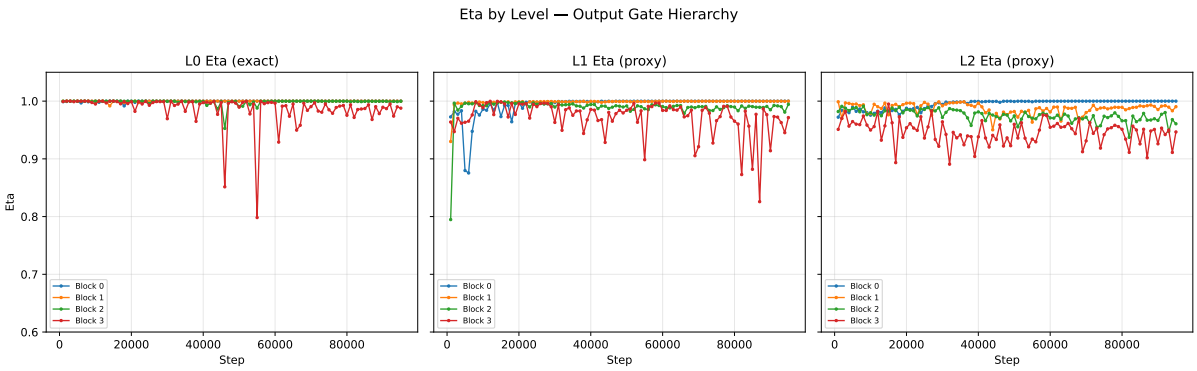

Figure 11: Eta by level. L0 saturates to 1.0. L1 and L2 both maintain high eta (>0.93).

Table 8: Eta at step 95K.

Block	**L0**	**L1**	**L2**
Block 0	1.000	1.000	1.000
Block 1	1.000	1.000	0.990
Block 2	1.000	0.994	0.961
Block 3	0.988	0.972	0.947

Critically, L1 eta does not collapse at k=3. In the k=2 130K run, Block 0's L1 eta dropped to 0.44–0.57 after step 57K. Here at 95K, Block 0's L1 eta is 1.000. Two possible explanations:

1. **L2 as pressure relief**: The presence of L2 below L1 may reduce the gradient pressure that caused L1 eta collapse at k=2. With three levels, the output gradient is distributed across L0/L1/L2 rather than concentrated on L0/L1.

2. **Different training phase**: This run starts from a k=2 checkpoint (pre-trained L1) rather than from scratch. The L1 gates may have settled into a more stable configuration before L2 was added.

4.4 DGD Delta Norms: L0 Only

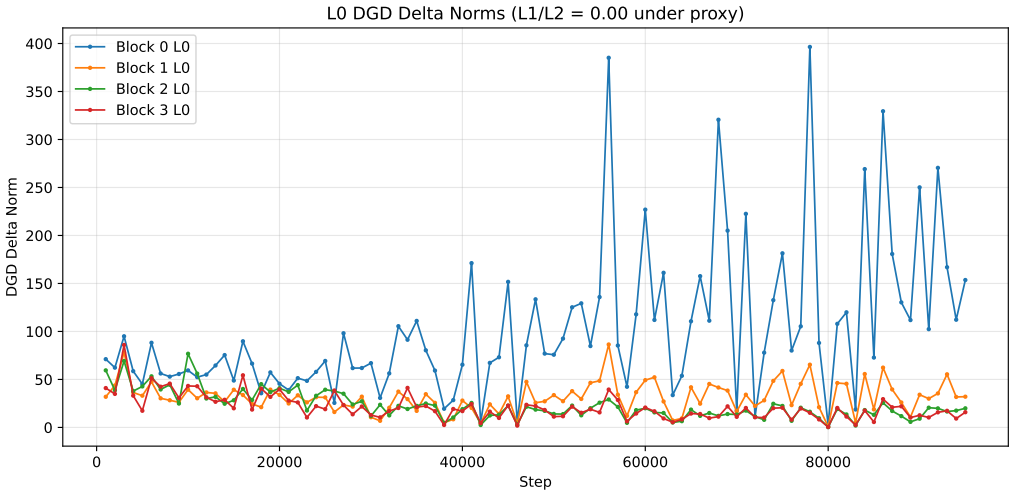

Figure 12: L0 DGD delta norms (L1/L2 read 0.00 under proxy). Block 0 L0 DGD grows from 71 to 154, indicating increasing L0 inner-loop update magnitude over time.

As with the k=2 proxy runs, DGD delta norm is only available for L0 (the exact-gradient level). L1 and L2 DGD norms are 0.00 because the proxy backward does not walk the M trajectory. Block 0 L0 DGD growth (71→154) is notable — the L0 inner-loop updates are getting *larger* over time, consistent with the model routing more learning through L0.

4.5 Block 2: The Persistent Anomaly

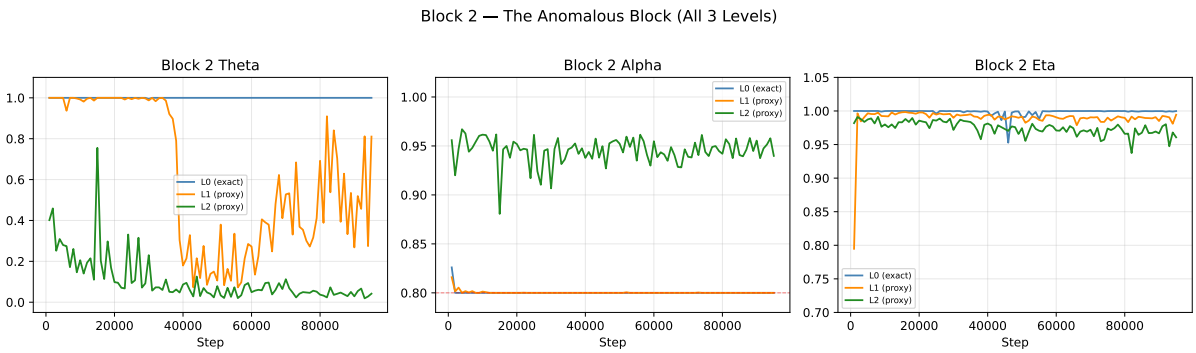

Figure 13: Block 2 gate detail across all three levels. L1 theta rises to 0.81 (similar to k=2 130K's 0.88). L2 theta is the highest of any block at 0.04. Alpha pinned to floor for L0/L1, elevated for L2 (0.94).

Block 2 continues to be the anomalous block:

- **k=2 30K report**: Block 2 L1 theta = 0.41 (only block with high proxy theta)
- **k=2 130K report**: Block 2 L1 theta = 0.88 (intensified)
- **d=1024 shakedown**: Block 2 L1 theta = 0.51 (persists at scale)
- **k=3 push-up (this run)**: Block 2 L1 theta = 0.81, L2 theta = 0.04 (highest of any block)

This is a robust, reproducible property of Block 2's position in the 4-block architecture. Block 2 receives sufficient gradient signal through the output path to sustain proxy-level learning at every scale and configuration tested.

5 Output Gradient Norms

Table 9: Output gradient norms at step 95K (identical within each block under MAG composition).

Block	L0	L1	L2
Block 0	0.0347	0.0347	0.0347
Block 1	0.0163	0.0163	0.0163
Block 2	0.0150	0.0150	0.0150
Block 3	0.0104	0.0104	0.0104

Output gradient norms are identical across all three levels within each block, confirming the MAG composition's structural property: all levels receive the same output gradient. The differentiation between levels comes entirely from the *backward pass through M*, not from the output gradient path. This is why proxy levels (which truncate the M backward) develop different gate dynamics than exact levels despite seeing the same output signal.

6 Throughput

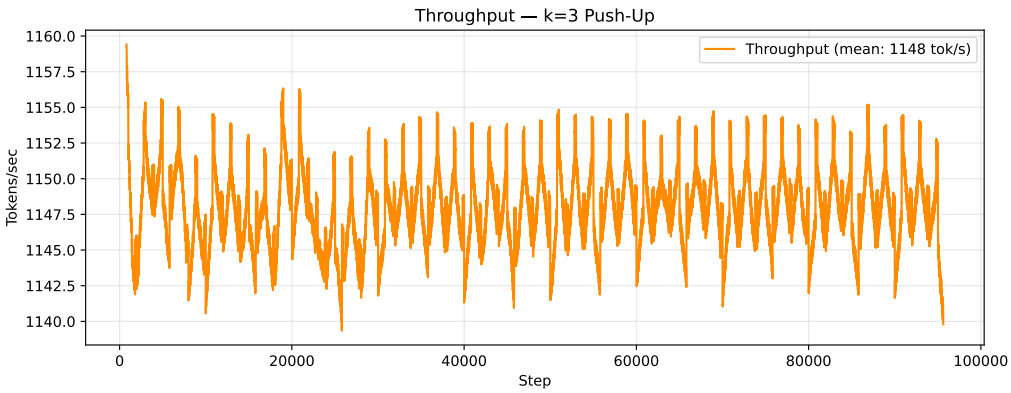

Figure 14: Throughput over 96K steps. Mean: 1,148 tok/s.

Table 10: Throughput comparison across runs.

Run	Tok/s	vs k=2 spec25
spec25 k=2 (all exact)	1,207	baseline
spec27 k=2 (L1 proxy, 130K)	1,186	−1.7%
k=3 push-up (this run)	**1,148**	**−4.9%**

Adding the third level costs ∼5% throughput versus k=2 all-exact. The cost is modest because L2 (chunk_size=64) fires rarely — only every 64th token. With L1 and L2 both using proxy backward (no M trajectory storage), the tape memory overhead is dominated by L0 alone. Wall time: 11.88h for 96K steps.

7 Conclusions

7.1 Push-Up to k=3 is Viable

The push-up from k=2 to k=3 produces a functioning three-level CMS that:

- Recovers from the push-up spike (3.84 → 2.53 in 50K steps).
- Achieves a better minimum smoothed loss (2.53) than the k=2 130K run (2.60).
- Maintains stable training with fewer gradient spikes per step than k=2.
- Adds only 5% throughput overhead.

However, the late-training oscillation (smoothed loss oscillating 2.5–2.8 after step 50K rather than converging) suggests the model may benefit from a longer schedule or different LR decay.

7.2 Three-Tier Hierarchy: Exact > Inherited Proxy > Fresh Proxy

The gate dynamics reveal a clear hierarchy determined by training history:

1. **L0 (exact gradients, trained from scratch)**: $\theta \approx 1.0$, $\eta \approx 1.0$. Dominant active learner.
2. **L1 (proxy gradients, inherited from k=2)**: θ bimodal — high in 3/4 blocks (inherited from exact-gradient phase), suppressed in Block 0. Active in blocks where it was pre-trained.
3. **L2 (proxy gradients, fresh initialization)**: $\theta < 0.05$ everywhere. Never bootstraps to meaningful learning. Passive memory retainer.

This hierarchy has a clear implication: **proxy gradients cannot bootstrap a level from scratch to active learning**. Levels must be pre-trained under exact gradients (or inherited from a phase that used exact gradients) before transitioning to proxy. The push-up pipeline inherently provides this: each level gets its exact-gradient phase before the next level is added and the previous level transitions to proxy.

7.3 No L1 Eta Collapse: The Pressure Relief Hypothesis

Unlike the k=2 130K run (Block 0 L1 eta collapsed to ∼0.48), L1 eta remains at 1.0 in Block 0 at k=3. This is the most significant structural difference between k=2 extended and k=3 push-up training. The hypothesis: with three levels sharing the output gradient, the pressure on any single level's output gate is reduced. L2's presence — even as a passive retainer — distributes the gradient load that caused L1 eta collapse at k=2.

7.4 Alpha Floor is Load-Bearing at k=3

With 10 of 12 block/level combinations hitting the alpha floor (0.80), the floor constraint is even more critical at k=3 than at k=2. Without it, the model would push retention below 0.80, causing faster forgetting that could destabilize memory across three levels simultaneously.

7.5 Next Steps

1. **Push-up to k=4**: Add Level 3 (chunk_size=512) from this checkpoint, completing the full CMS hierarchy. tape_strategies=[exact, proxy, proxy, proxy].

2. **Longer cosine schedule**: The late-training oscillation may resolve with a 200K+ schedule instead of 100K, giving the three levels more time to equilibrate.

3. **L2 exact-gradient ablation**: Test whether giving L2 exact gradients (instead of proxy) allows it to bootstrap to higher theta. If so, this confirms the hierarchy is gradient-quality–driven, not structural.

4. **Shard M diff logging**: Needed for L1/L2 differentiation diagnostics (DGD unavailable under proxy).